

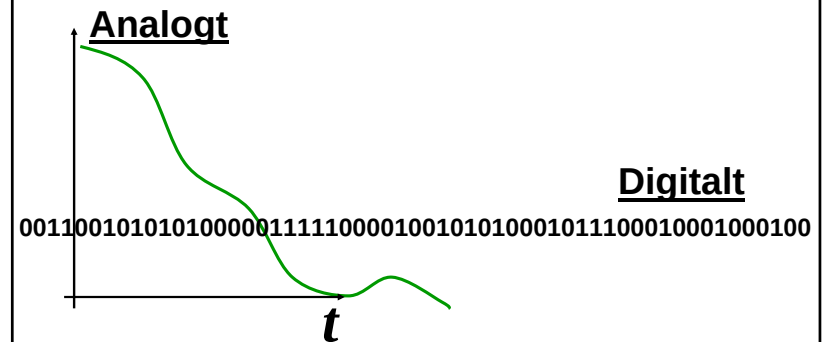
Analogt och Digitalt

Viktor Öwall



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Analoga och Digitala Signaler



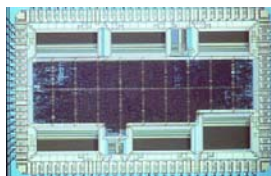
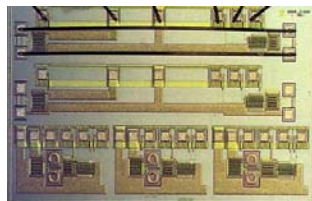
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Analogt kontra Digitalt

Analogt

- få komponenter
- låg effektförbrukning

"verkliga" signaler

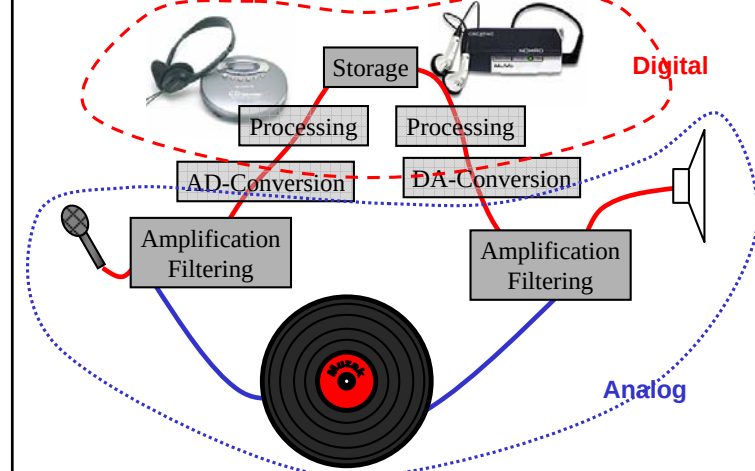


Digitalt

- Hög precision
- Komplexare algoritmer
- Lagringskapacitet

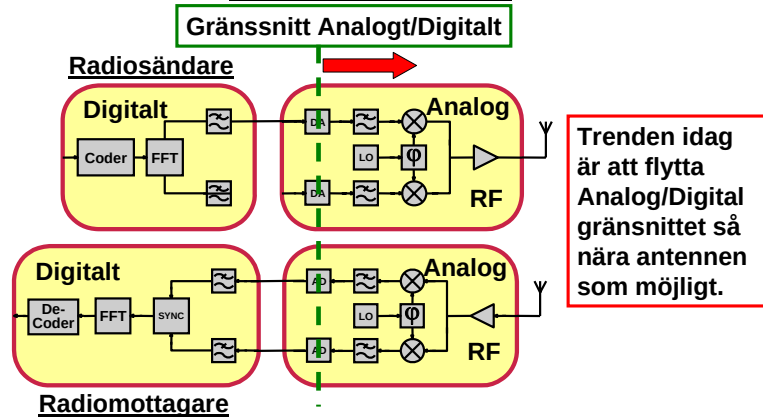
CD/DVD, MP3, Digitalkamera, GSM, datorer, etc, etc

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AD/DA i Radio



Binära Talsystemet

MSB =
Most Significant Bit

LSB =
Least Significant Bit
Minsta förändringen

	2^2	2^1	2^0	
0	0	0	0	(0)
0	0	1	1	(1)
0	1	0	0	(2)
0	1	1	1	(3)
1	0	0	0	(4)
1	0	1	1	(5)
1	1	0	0	(6)
1	1	1	1	(7)

N bitar
↓
 2^N ord

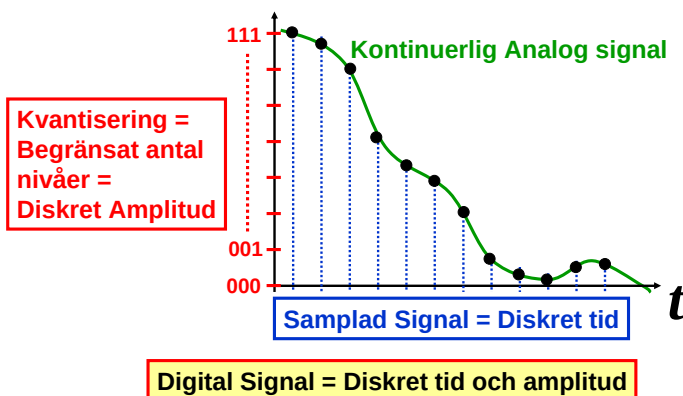
$$V_{fs} =$$

$$V_{full\ scale} = 1LSB \cdot 2^N$$

(Med denna definition kan inte V_{fs} nås)

$$V_{outmax} = V_{LSB} (2^N - 1)$$

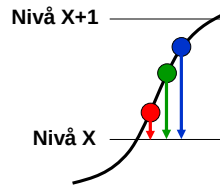
Digitala Signaler



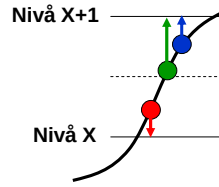
Dynamik och Upplösning

Antal bitar	Antal Intervall	Upplösning $V_{fs}=0.5V$	Dynamik $V_{LSB}=0.03125$
4	16	0.03125V	0.5V
8	256	2mV	8V
12	4096	0.12mV	128V
16	65 536	7.6 μ V	2042V

Kvantiseringsfel



Trunkering
Alla värden mellan två nivåer approximeras åt samma håll
Maximalt fel = 1LSB



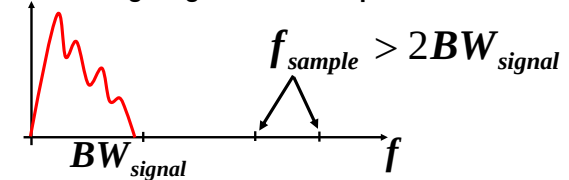
Avrundning
Värden approximeras antingen upp eller ner
Maximalt fel = 1/2 LSB

Nyquists Samplingsteorem

Om man samplar en analog signal med en bandbredd, BW_{signal} , med en samplingsfrekvens

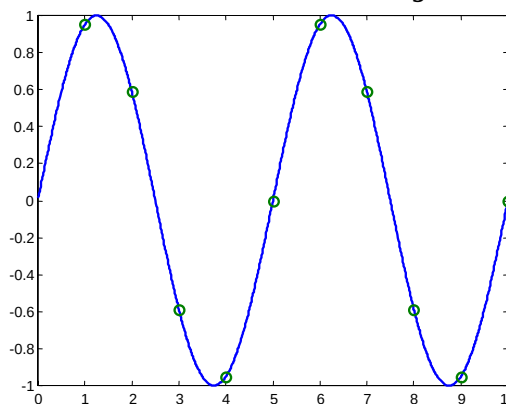
$$f_{sample} > 2BW_{signal}$$

kan den analoga signalen återskapas.

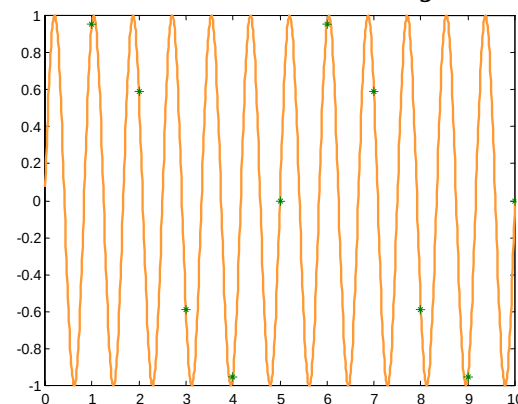


Mer om detta i Digital Signalbehandling

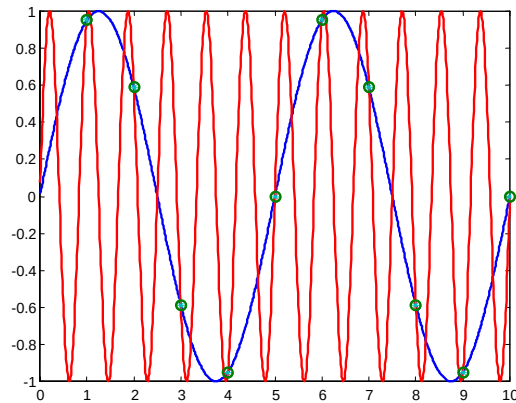
Sampling - $f_s > 2f_{signal}$



Sampling - $f_s < 2f_{signal}$

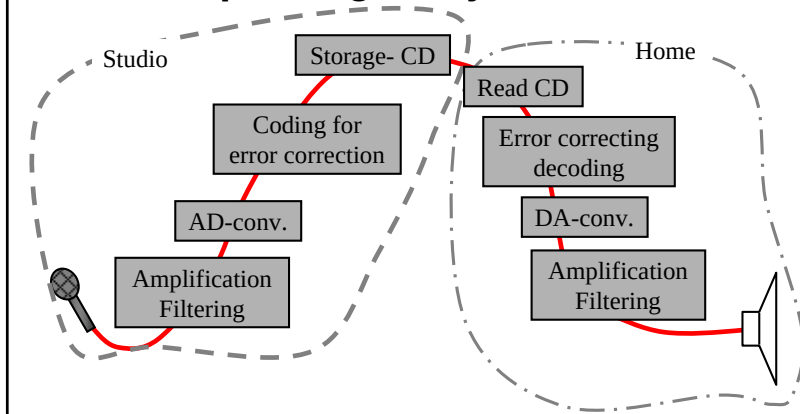


Signalerna kan förväxlas - vikning



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Exempel: Digitalt ljud, CD/DVD



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CD kontra DVD audioformat

Specification	CD Audio	DVD Audio
Sampling Rate	44.1 kHz	96/192 kHz
Sampling Accuracy	16-bit	24-bit
Number of Possible Output Levels	65,536	16,777,216

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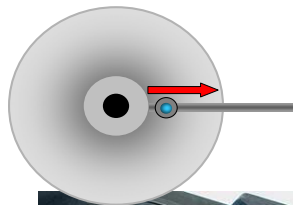
CD/DVD

Om man sträcker ut spåret på en CD blir det nästan 5 km.



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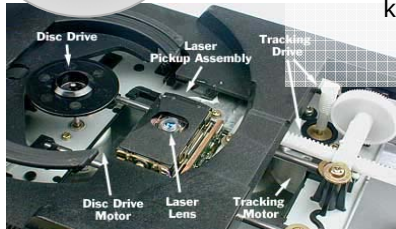
CD spelaren



Lasern flyttas från centrum och utåt och hastigheten reduceras från 500rpm to 200 rpm



"The bumps" passerar lasern med konstant hastighet



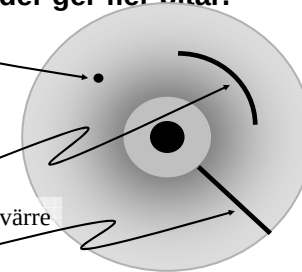
Felkorrigering koder

Data kodas så om ett "begränsat" antal fel uppstår kan de korrigeras.

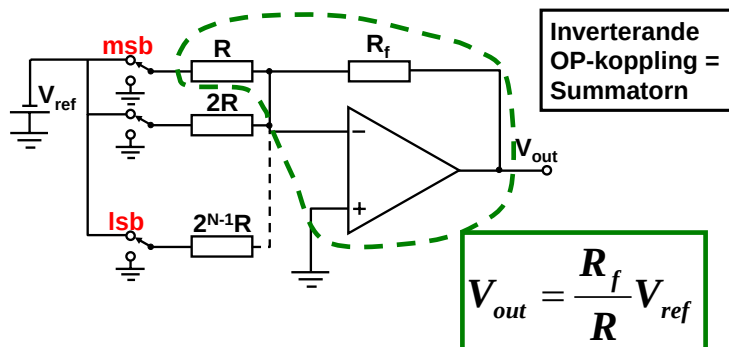
Felkorrigering koder ger fler bitar.

Man kan borrar ett "litet" hål och fortfarande spela CDn.

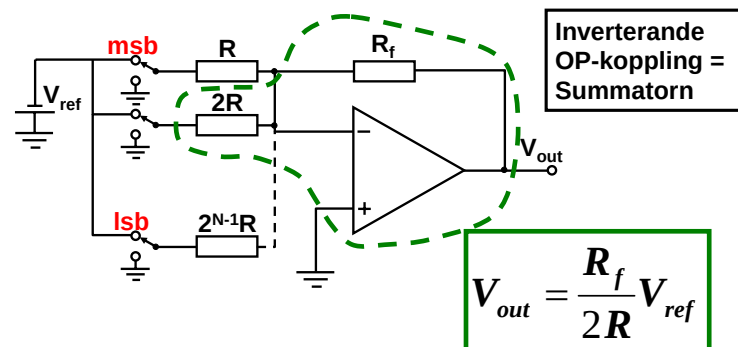
"fel" längs spåren är värre än diagonala.



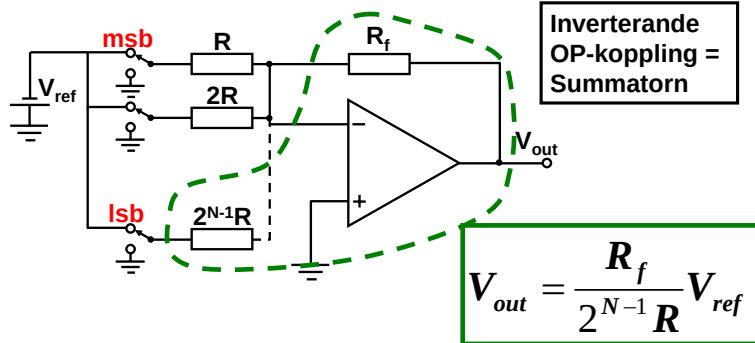
Digital till Analog Konvertering



Digital till Analog Konvertering

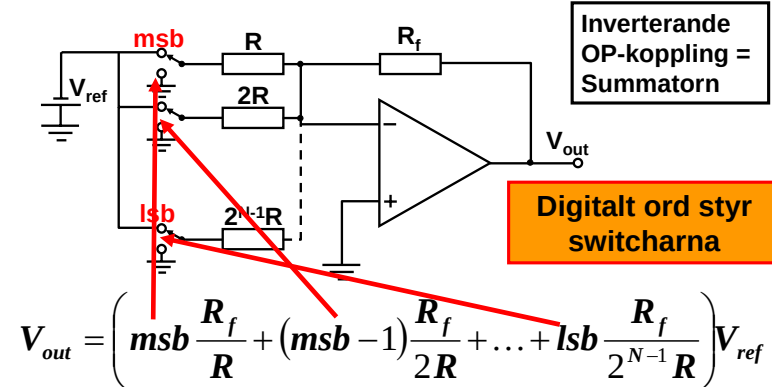


Digital till Analog Konvertering



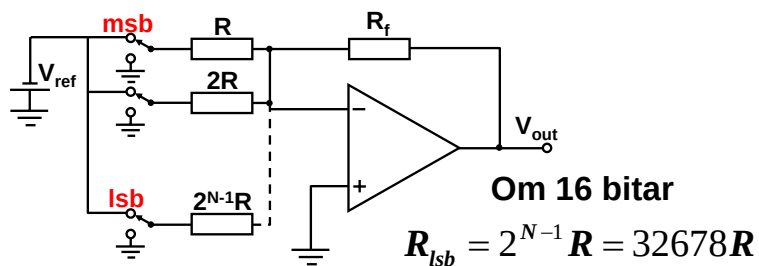
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Digital till Analog Konvertering



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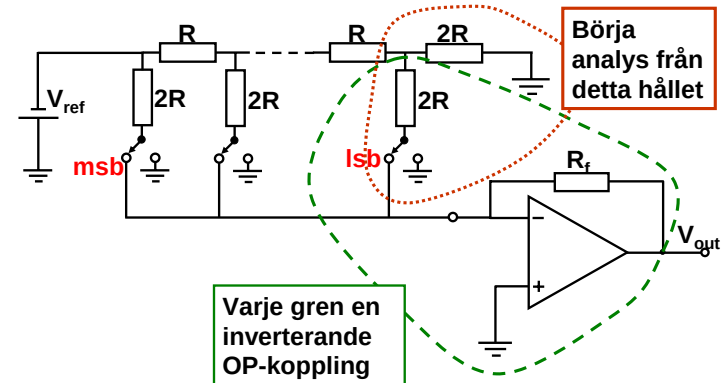
Digital till Analog Konvertering



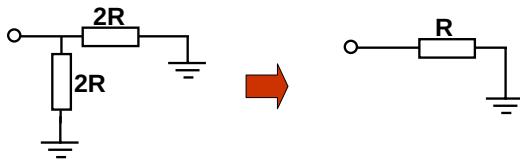
**Svårt med så stora motståndsvärden!
Litet antal bitar med relativt låga krav.**

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DA-omvandling - R/2R Stege

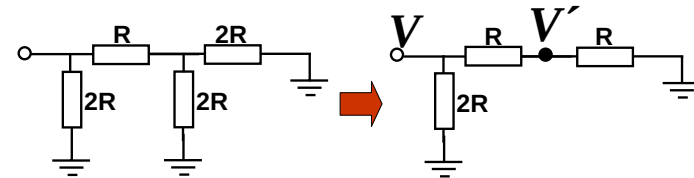


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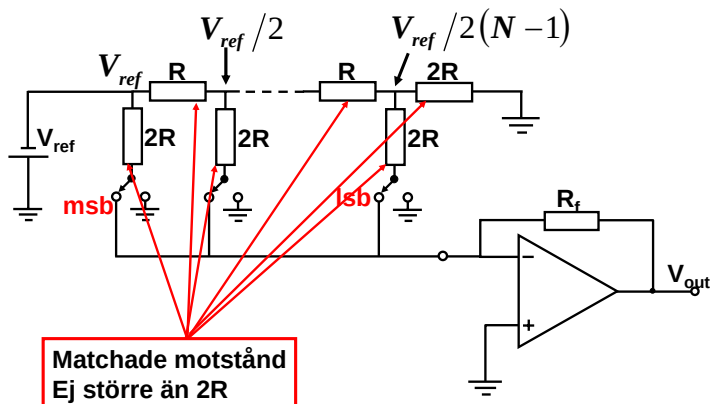
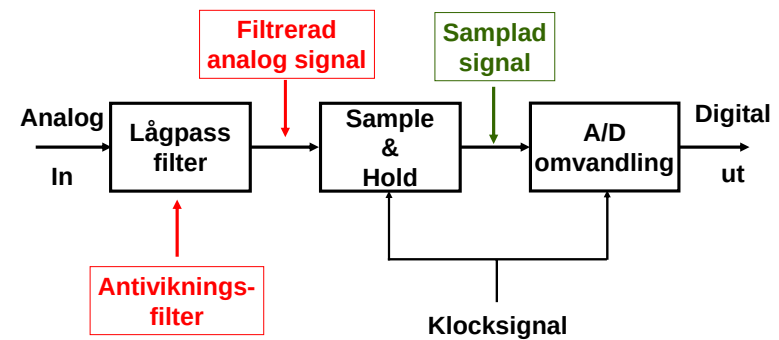
DA-omvandling - R/2R Stege

Parallellkoppling av motstånd

$$\frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

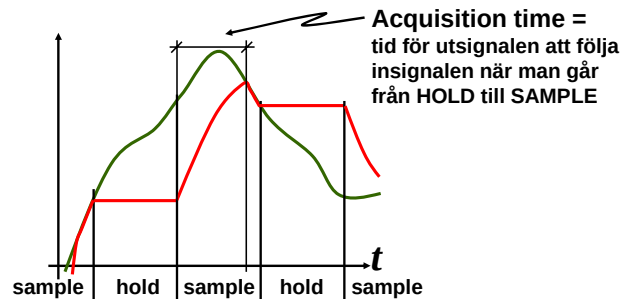
DA-omvandling - R/2R StegeSpänningshalvering
i varje steg

$$V' = \frac{R}{R+R} V = \frac{V}{2}$$

DA-omvandling - R/2R Stege**Analog till Digital omvandling**

Sample & Hold

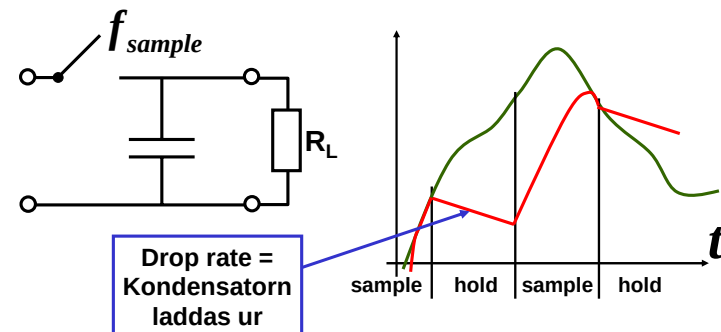
Under AD-omvandlingen får inte det analoga värdet ändras → Sample & Hold krets



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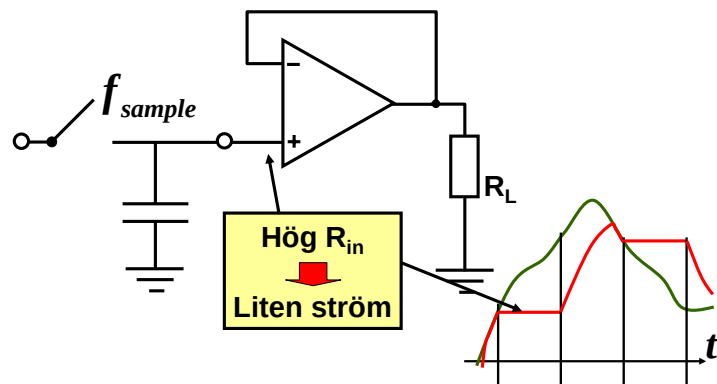
Sample & Hold

Spara ett analogtvärde → Kondensator



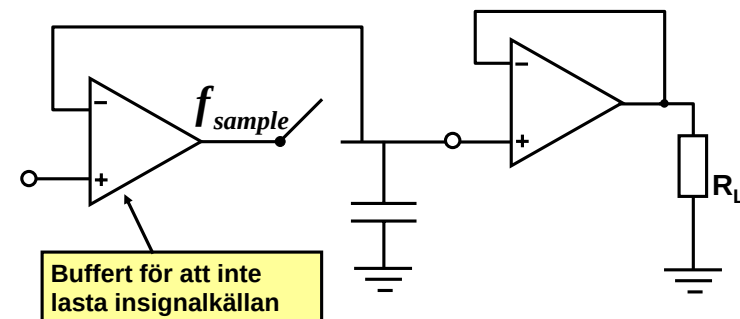
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Sample & Hold med Buffer



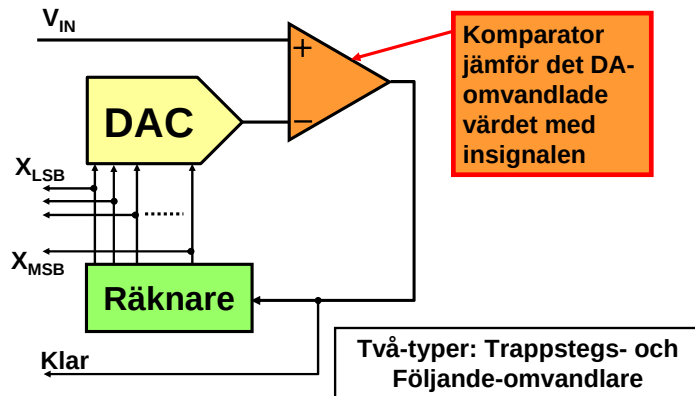
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Sample & Hold med Buffer



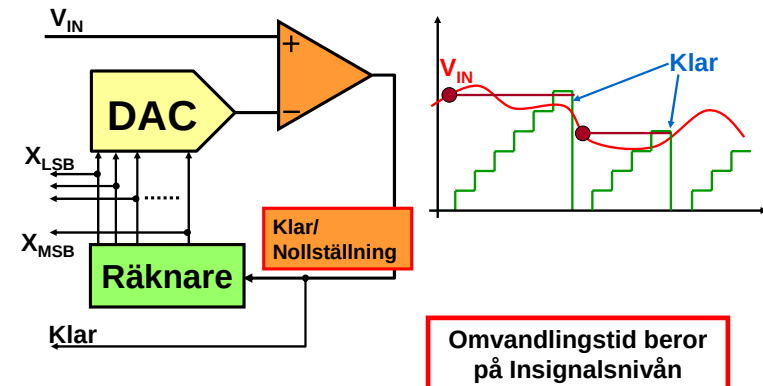
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Räknarbaserad AD-omvandlare



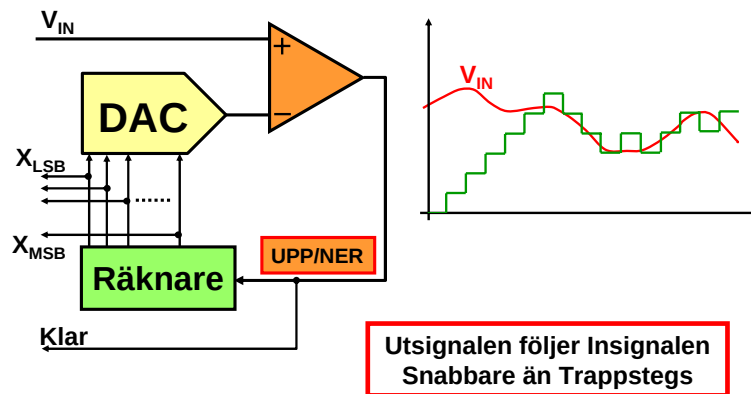
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Trappstegsomvandlare



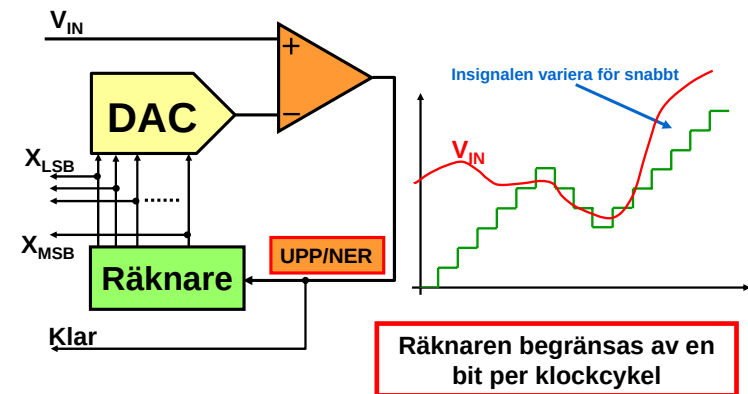
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Följandeomvandlare



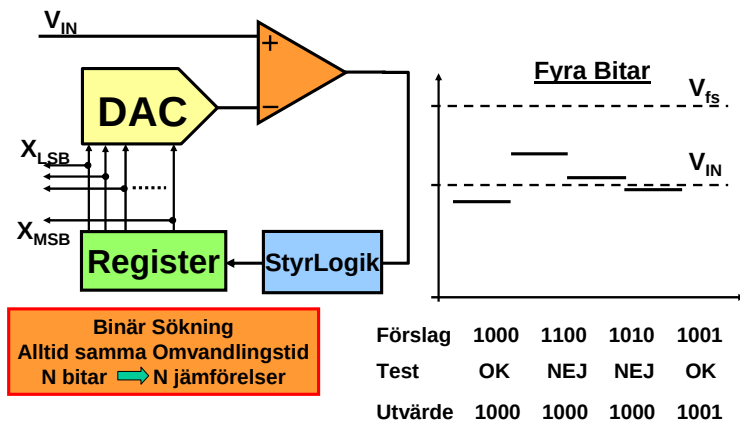
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Följandeomvandlare



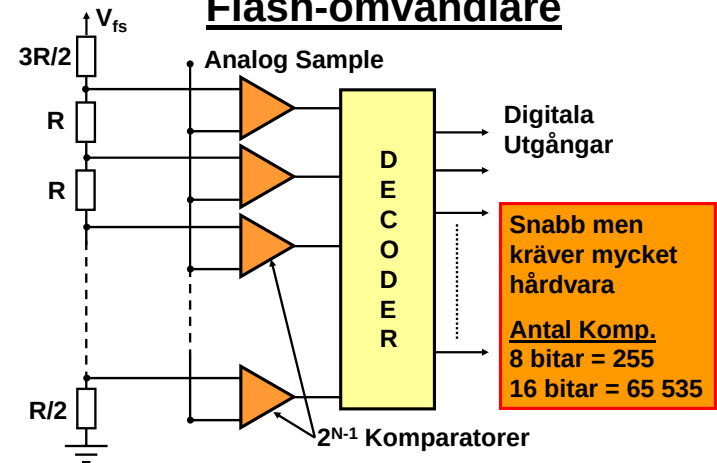
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Succesivapproximation



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Flash-omvandlare



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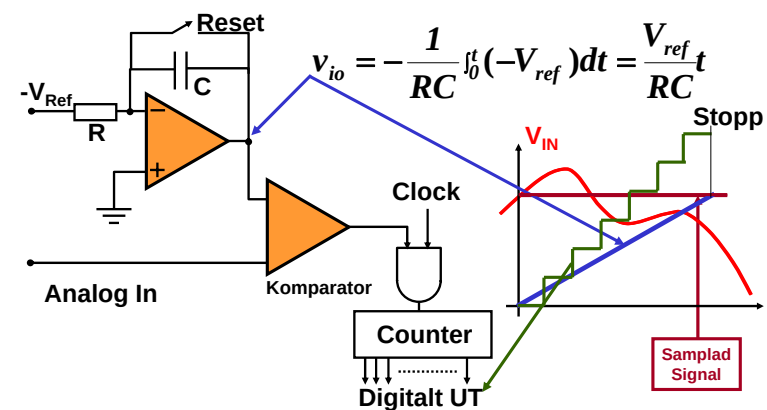
Omvandlingstid

Trappstegsomvandlare	$2^N - 1$
Succesiv approximation	N
Flash	1

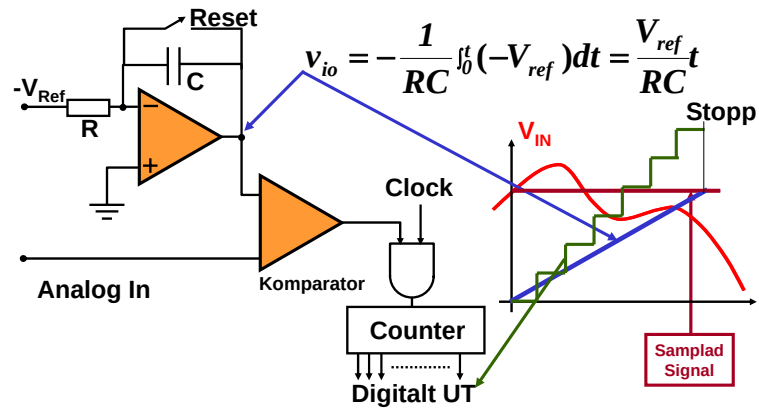
Men snabbare omvandling
betalas med extra hårdvara

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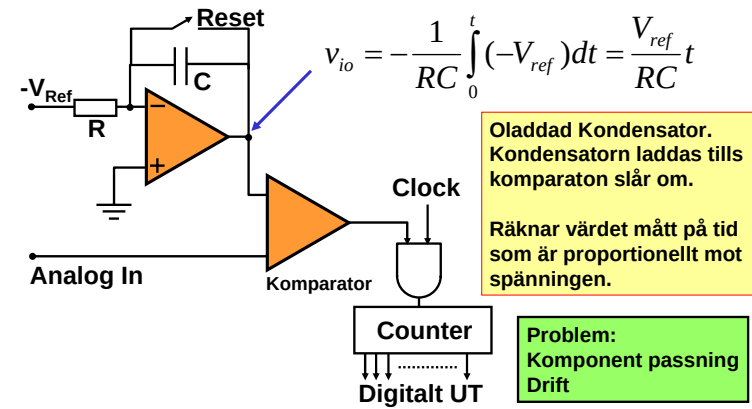
Integrerande Omvandlare, Single Slope



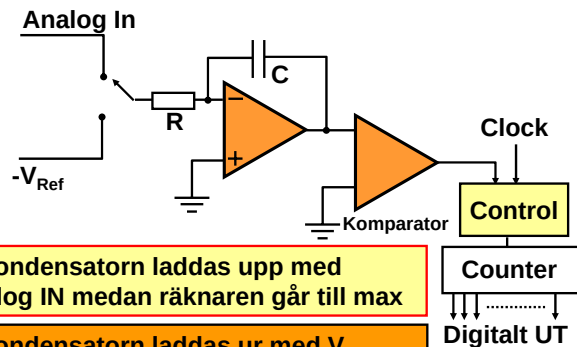
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Integrerande Omvandlare, Single Slope

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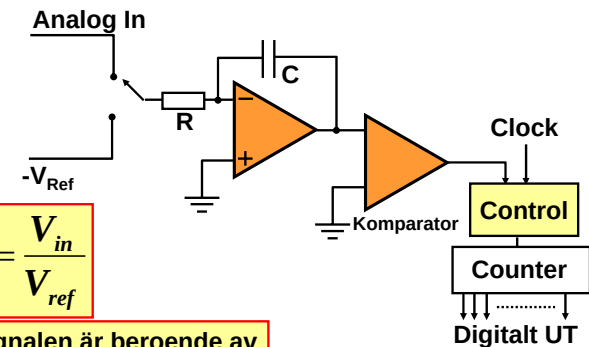
Integrerande Omvandlare, Single Slope

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Integrerande Omvandlare, Dual Slope

1. Kondensatorn laddas upp med Analog IN medan räknaren går till max
2. Kondensatorn laddas ur med V_{ref} , räknar värdet vid omslag mått på analog spänningen

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Integrerande Omvandlare, Dual Slope

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